MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)
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MR-22

B.Tech./M.Tech. – SEMESTER - I
MID-TERM EXAMINATION - I
PART B: DESCRIPTIVE EXAMINATION

Course **: Environmental Science** Time **: 90 Min**
Date & Session **: DD/MM/YYYY** Maximum Marks **: 20**
Branch & Section**: CSE, IT, CSB.CSD.CSM** Roll No**:**

Answer Key

Note: Answer any FOUR full questions. All questions carry equal marks. **(4X5M=20 Marks)**

Q.No.	Stem of the Question	M	L	CO
1.a)	**Describe the structure of ecosystem.**	3	1	1

Ans: Structure of ecosystem:

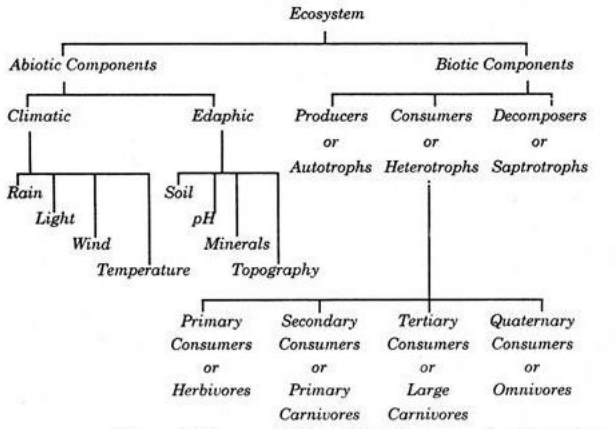
The structure of an ecosystem can be split into two main components, namely:

Biotic Components and Abiotic Components.

Biotic structure includes plants, animals & microorganisms present in an ecosystem.

These organisms have different nutritional behavior and status in the ecosystem and are known as **Autotrophs** (Producers), **Heterotrophs** (Consumers) & **Micro-consumers** (Decomposers) based on how they get their food.

Abiotic components are the non-living component of an ecosystem. It includes air, water, soil, minerals, sunlight, temperature, nutrients, wind, altitude, turbidity, etc.



Schematic Representation of the Structure of an Ecosystem.

b)	**Write the impact of overutilization of groundwater.**	2	4	2

Ans. Impact of overutilization of groundwater.

Over use of groundwater has following effects……

1. Lowering of water table:

Excessive use of ground water for drinking, irrigation and domestic purposes has resulted in rapid depletion of ground water in various regions leading to lowering of water table.

2. **Ground subsidence**

When ground water withdrawal is greater than its recharge rate, the sediments in the aquifer become compacted. This is called ground subsidence which may cause damage of buildings, destroy water supply systems etc.

3. **Intrusion of salt water**

In coastal area, over exploitation of ground water leads to rapid intrusion of salt water from

the sea. As a result, water cannot be used for drinking and agriculture.

4. **Earthquake and landslides**

Over-utilization of ground water leads to decrease in water level causing earth quake and landslides.

5. **Drying of well**

Excess withdrawal of ground water, result depletion of ground water level and drying up of the well.

6. **Pollution of water:**

When the gr water level near agricultural land decreases, water containing fertilizers percolate rapidly into the aquifer and pollute ground water. As a result water becomes unsuitable for drinking.

2.a) Why the pyramid of energy always takes upright shape? 3 2 1

Ans: Pyramid of Energy

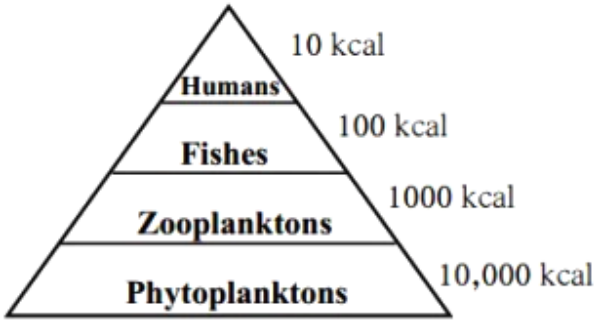
It shows the energy available at each trophic level.

The size of each trophic level represents the proportion of productivity

It is measured in Joules or Calories.

Most of the energy available to the community is in the 1st trophic level.

Only 10-20% of the energy is available to the next trophic level (≈ 90% lost)



Pyramid of energy in an aquatic ecosystem

Pyramid of **energy** is the only type of **ecological pyramid**, which is **always upright,** because the **energy** flow in a food chain is **always unidirectional**.

Also, with every increasing trophic level, some **energy** is lost into the environment as heat energy.

So if we represent any ecosystem food chain in terms of the energy, the value will always decrease from producer to higher trophic levels.

Thus pyramid shape turns upright only.

b) Enlist various causes of deforestation. 2 3 2

Ans. **Causes of deforestation**

(a) Development and construction works
(b) Mining operation
(c) Various raw materials demand for industry
(d) Fuel requirement for villager and tribal people
(e) Shifting cultivation
(f) Forest fire etc.

3.a) How N_2 cycle maintain proper functioning of the ecosystem? 2 1 1

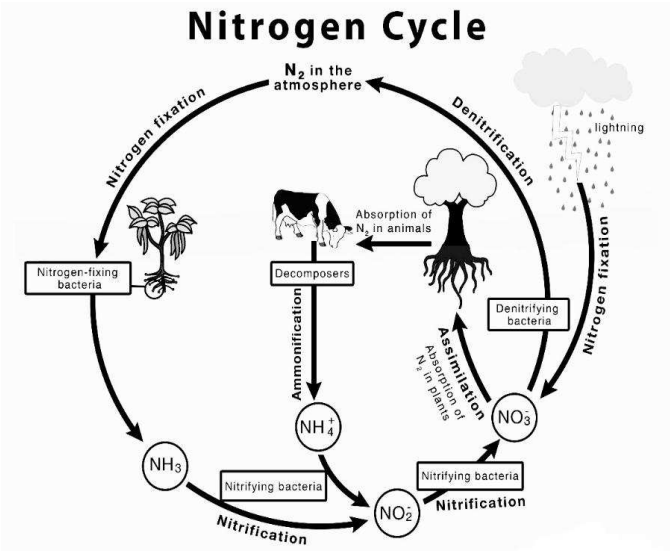
Ans. **<u>Nitrogen cycle</u>**

<u>Nitrogen cycle</u> is that biogeochemical cycle through which the movement of nitrogen occurs between biotic (living) and abiotic (lithosphere, hydrosphere, atmosphere) components of the ecosystem.

Process of Nitrogen Cycle consists of the following steps ..

1. **Nitrogen fixation:** Nitrogen from the atmosphere is converted into ammonia and other nitrogeneous compound by certain bacterial species like Rhizobium, Azotobacter, etc. and by other natural phenomena.
1. **Assimilation:** Nitrogen in the form of nitrite ions, nitrate ions or ammonium ions enters in the plant and animal body.
2. **Ammonification:** After the death of plants and animals nitrogen present in organic matter converts to ammonium compounds by ammonifying bacteria,
3. **Nitrification:** Nitrites and nitrate salts are formed by the oxidation of ammonia with the help of <u>Nitrosomonas</u> bacteria species.
4. **Denitrification:** Inorganic N_2 is produced from nitrate /nitrite salts by denitrifying bacteriaand released into the atmosphere

A schematic representation of the whole cycle is given here….



b) What is drought? Write the impacts of drought. 3 2 2

Ans. Drought: A **drought** is a natural phenomenonn of prolonged shortages in the water supply, whether atmospheric, surface water or ground water.

Impacts	Impacts of drought				
	Agriculture Food security	**Water Sanitation**	**Health**	**Infrastructure Shelter**	**Economic**
Direct	Loss of agricultural output	Deterioration of water quality	Health status (e.g. malnutrition)	Wells	Funds diverted to relief
	Loss of livestock	Deterioration of sanitary conditions		Water distribution	Business loan defaults
	Food prices	River hydrology (decreased flow)		Irrigation	Local spending
	Reduction of demand for labour				
	Food shortage				
Indirect	Household assets liquidated	Fetching water	Morbidity		Government revenue, ForEx
	Migration	Water disputes	Mortality	Loss of shelter	Inflation
	School attendance				
	Land degradation and declining productivity				Diminished spending

4.a) Summarize the major environmental effects of over-exploitation of minerals. 3 2 2

Ans. **Impact on environment due to over-utilization of minerals**

(i) Devegetation and defacing of landscape

- Topsoil and vegetation is removed from mining area to get assess of minerals
- Large scale deforestation or Devegetation leads to ecological loss
- Landscape get badly affected
- Huge quantities of debris and alongwith big scar disrupt the aesthetic value of the area.

(ii) Subsidence of land

- Associated with underground mining

It results in –

Tilting of buildings

Cracks in houses

Buckling of roads

Bending of rail tracks

Leaking of gas from cracked gas pipelines

(iii) Groundwater contamination

- Mining pollutes groundwater
- Affects hydrological cycle
- Sulphur normally present as impurity in ore get converted into sulphuric acid through microbial action
- Some heavy metals leached into groundwater contaminated it
- Posing health hazards

(iv) Surface water pollution

- Acid mine drainage often contaminates the nearby streams and lakes.
- The acidic water detrimental to aquatic life
- Radioactive material affects the aquatic life
- Heavy metals contaminates the water body and affects the aquatic life

(v) Air pollution

- Smelting is done to purify the metals from other impurities
- During smelting enormous quantites of air pollutant released affects

Vegetation

Serious environmental impacts

Eg SPM, soot, lead ,cadmium shoot up in smelter atmosphere causes health problems

(vi) Occupational Health Hazards

- Miners suffers from respiratory and skin problems due to constant exposure to SPM and other toxic

- Miners working in various mining area suffer from silicosis (inhaling silica dust), asbestosis (due to asbestos fibres), black lung disease (inhaling coal dust), etc.

b) Differentiate biomagnification and bioaccumulation. 2 5 1

Ans.

BIOACCUMULAION	BIOLOGICAL MAGNIFICATION
The gradual accumulation of substances such as pesticides or other chemicals in living organisms	The concentration of a toxic chemical in the tissues of tolerant organisms at the successively higher levels in a food chain
The process of building-up of toxic chemicals in the body of a living organism	The increase of the toxic chemicals in the amount each time they move up a food chain
Caused by the uptake of toxic chemicals into the body either through the skin or through ingestion	Toxic chemicals start in small amounts at the lower levels of the food chain and gradually increase in amount when it moves to the higher levels
Results in chronic poisoning even when the concentration of the toxic compound in the surrounding environment is not too high	Results in higher concentrations of toxic substances at the higher tropic levels of a food chain

5.a) Define biodiversity. Give various levels of biodiversity. 3 3 3

Ans. Biodiversity is the variety and variability of life on Earth. It means "the existence of a large number of different kinds of animals and plants which make a balanced environment"

Biodiversity is usually considered at three different levels…….

1. **Genetic diversity**

The variety of genes within a given species

Genetic diversity is the diversity within species ie., variation of genes within the species. eg.,rice varieties.

teak wood trees : Indian teak,burma teak,malasian teak.



2. **Species diversity**

The variety of species in a given ecosystem.

Diversity is greatest when all the species present are equally abundant in the area.

A large number of plants, animals and microorganisms interact with each other in a ecosystem.



3. Ecosystem diversity

The variety of ecosystem in a given region

Ecosystem diversity deals with the variations in ecosystems within a geographical location and its overall impact on human existence and the environment.

An example of ecological diversity on a global scale would be the variation in ecosystems, such as deserts, forests, grasslands, wetlands and oceans.



b) Why food web is more important ecologically than the concept of simple food chain. 2 2 1

Ans. The transfer of food energy from the producers (plants) through a series of organisms (Herbivores, Carnivores) successively with the repeated activities of eating and being eaten is known as **food chain**.

Whereas **food web** is a network of food chains where different types of organisms are connected at different trophic levels so that there are a number of options of eating and being eaten at each trophic level.

So, food webs improve the stability of an ecosystem as it includes all the interconnected food chains. These webs are important for illustrating the feeding relationships in a community among the species.

A food chain is a linear sequence of organisms through which food and its contained energy pass. If one of the organisms number decreases in a linear food chain, next organisms will be affected due to food shortage.

However, in a food web, each organism has a choice to feed on different types of other organisms. So, the concept of the food web is more significant that food chain.

6.a) Discuss various values of biodiversity. 3 1 3

Ans. Various values of biodiversity

(i) Consumptive values,

- The value of Nature's Products that are consumed directly such as firewoods , fodder and meat.
- In other words the products which are consumed directly without passing through the market
- Consumptive use value seldom appear in National income accounts.

(ii) Productive use value

Commercially usable values where the product is marketed and sold

- Wild Gene Resources – traded for use by scientists for introducing desirable traits in the crops and domesticated animals

Many industries are built depending on the productive used value of biodiversity. 2 2 2

For e.g. industries like paper, pulp, plywood, textile, leather, pearl etc.

(iii) Social value

Social values of biodiversity refers to the manner in which the bio-resources are used to the society.

- These are the values associated with the social life, religion and spiritual aspects of the people.
- Holy plants: Many of the plants are considered holy plants in our country like *Tulsi,* Mango, Lotus, Neem etc.
- Holy animals : Many of the plants are considered holy animals in our country like cow, snake, peacock etc.,

(iv) Ethical value

Ethical and moral value is based on the importance of protecting all forms of life.

It involves ethical issues like ‘’All life must be preserved’’

The ethical value means that a species may or may not be used, but its existence in our nature gives us pleasure.

Ex: 1. The river Ganga is considered as holy river.

2. Vembu, Tulsi are some of the trees worshipped by the Tamilans.

3. We are not deriving anything from Kangaroo, Zebra, but we feel that these should exist in nature.

(v) Aesthetic value

The beautiful nature of plants & animals insist us to protect the biodiversity.

Ex: People from far and wide spend a lot of time and money to visit beautiful areas where they can enjoy the aesthetic value of biodiversity and this type of tourism is now known as ecotourism.

The pleasant music of world birds, colour of flowers, peacock, and butterfly are important aesthetic value.

(vi) Option value

It includes the potential of biodiversity that are presently unknown and need to be explored. The optional values of biodiversity suggest that any species may be proved to be a valuable after someday. (i.e., presently unknown but need to be known)

Eg., growing Biotechnology field is searching a species which can cure diseases like cancer and AIDS.

b) What are alternative energy resources? How it fulfill the future energy need.

Ans. The energy resources that can be regenerated easily through natural process once they are used up are called the renewable or alternative energy resources.

Renewable energy resources include solar energy, wind power, hydropower etc.

As the conventional fossil fuel energies like coal, petroleum, natural gas and nuclear energy are not abundant and one day will be exhausted, so we should depend more on the alternative non-conventional energy resources. The advantages of alternative energies are

(i) They are plenty in sources

(ii) Non-polluting and eco-friendly

These types of energy can fulfill our future energy demand.

Various types of alternative energy resources are

- Solar energy
- Wind

• Hydropower • Tidal energy • Ocean thermal energy • Geothermal • Biomass • Hydrogen fuel	

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome